

1. Definitions

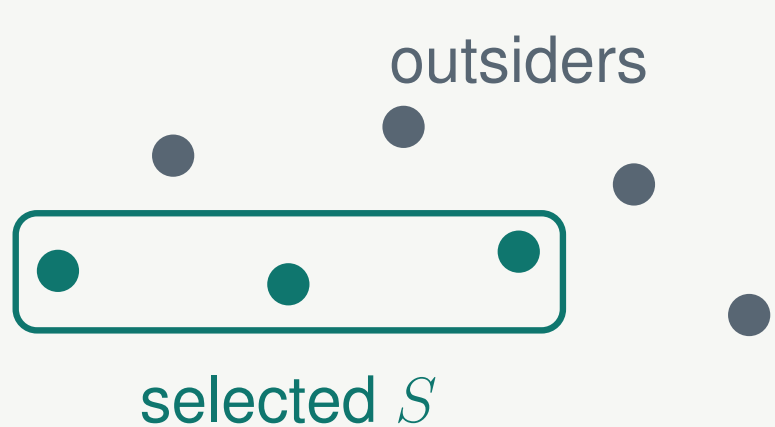
MED is a dimension of a score family, not a training protocol.

Store m objects as $X = \{v_1, \dots, v_m\} \subset \mathbb{R}^d$. For every answer set $S \subseteq X$ with $1 \leq |S| \leq k$, a query vector u_S and threshold b_S must retrieve exactly S .

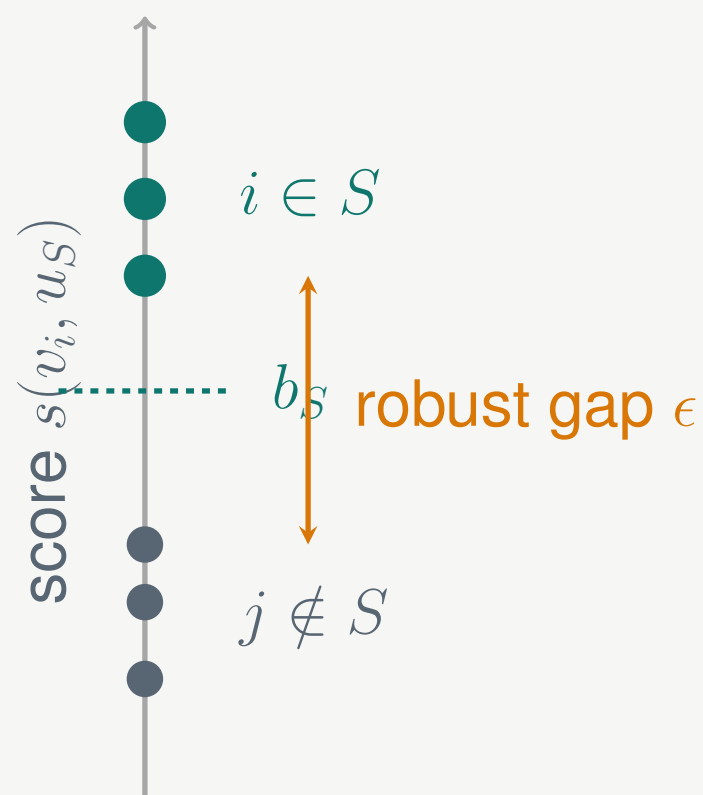
What must be realized?

Target set S

Score axis



Each subset up to size k may choose its own query vector. MED asks whether all such subsets can be realized in one object configuration.



Exact MED

$$\min_{i \in S} s(v_i, u_S) > b_S > \max_{j \notin S} s(v_j, u_S).$$

Strict ordering is enough. No prescribed margin or normalization is required.

Robust MED

$$\min_{i \in S} s(v_i, u_S) \geq \max_{j \notin S} s(v_j, u_S) + \epsilon.$$

Objects and queries are normalized. A uniform positive score gap must work for every target set.

VC lower bound

If all subsets of size at most k can be realized, then the thresholded score class shatters any chosen k objects. Thus

$$k \leq \text{VCD}(\mathcal{C}_{S,d}).$$

For inner-product and Euclidean threshold families, $\text{VCD} = d + 1$, so exact MED and every robust MED satisfy $d \geq k - 1$.

Lower bounds are $\Omega(k)$; the question is whether m must enter.

2. Theory

Exact capacity is m -independent; robust capacity pays for margin.

Exact MED theorem

Score	Lower	Upper
Inner product	$k - 1$	$2k$
Euclidean	$k - 1$	$2k$
Cosine	$k - 1$	$2k + 1$

Main message: $\text{MED}(m, k) = \Theta(k)$, independent of m up to constants.

Cyclic-polytopal witness

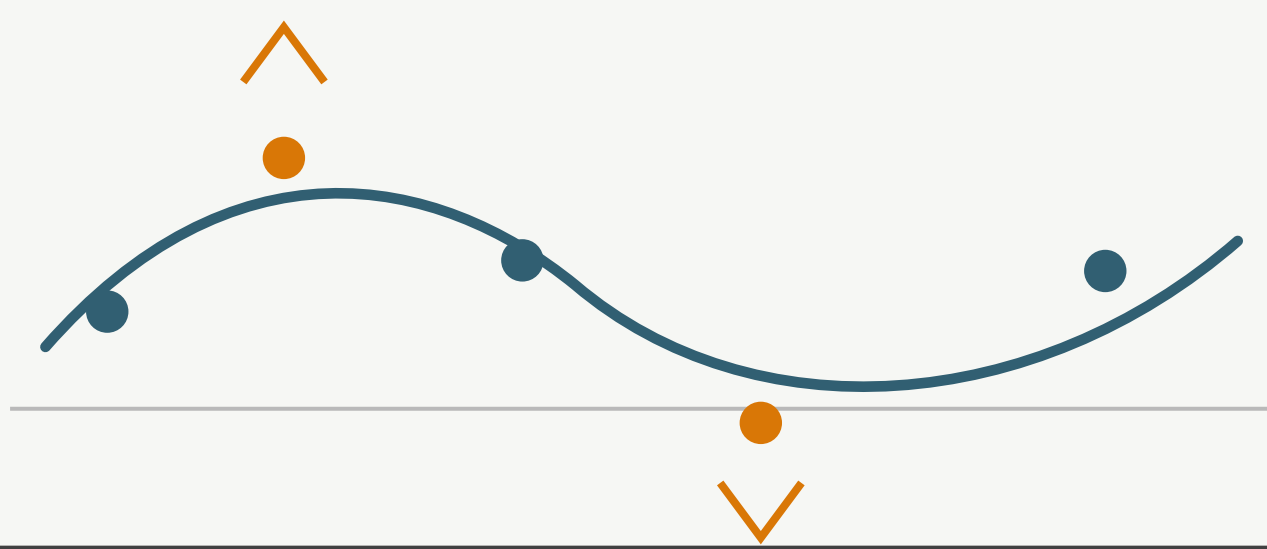
Put all m objects on the moment curve in $\mathbb{R}^{2k}$:

$$v_i = (t_i, t_i^2, \dots, t_i^{2k}).$$

For target S , use the square polynomial

$$P_S(t) = \prod_{i \in S} (t - t_i), \quad \langle v(t), q_S \rangle = c_0 - P_S(t)^2.$$

Selected points are exactly the roots of P_S and score c_0 ; every outsider scores below c_0 .



Feasible robust margin

$$\epsilon_*(m, k) = \frac{m}{\sqrt{k(m-1)(m-k)}} \sim \frac{1}{\sqrt{k}}.$$

If $\epsilon > \epsilon_*(m, k)$, robust retrieval is impossible in any finite dimension. The ceiling is tight in $\mathbb{R}^{m-1}$ by a regular simplex.

Gaussian centroid upper bound

Sample random unit object vectors and query by normalized selected centroid:

$$u_S = \frac{\sum_{i \in S} v_i}{\left\| \sum_{i \in S} v_i \right\|_2}.$$

Selected objects get a self-correlation term; outsiders contribute coherence noise. With positive probability,

$$d = Ck^2 \log m$$

robustly retrieves every at-most- k subset with margin $\Omega(1/\sqrt{k})$.

Score-rule transfer

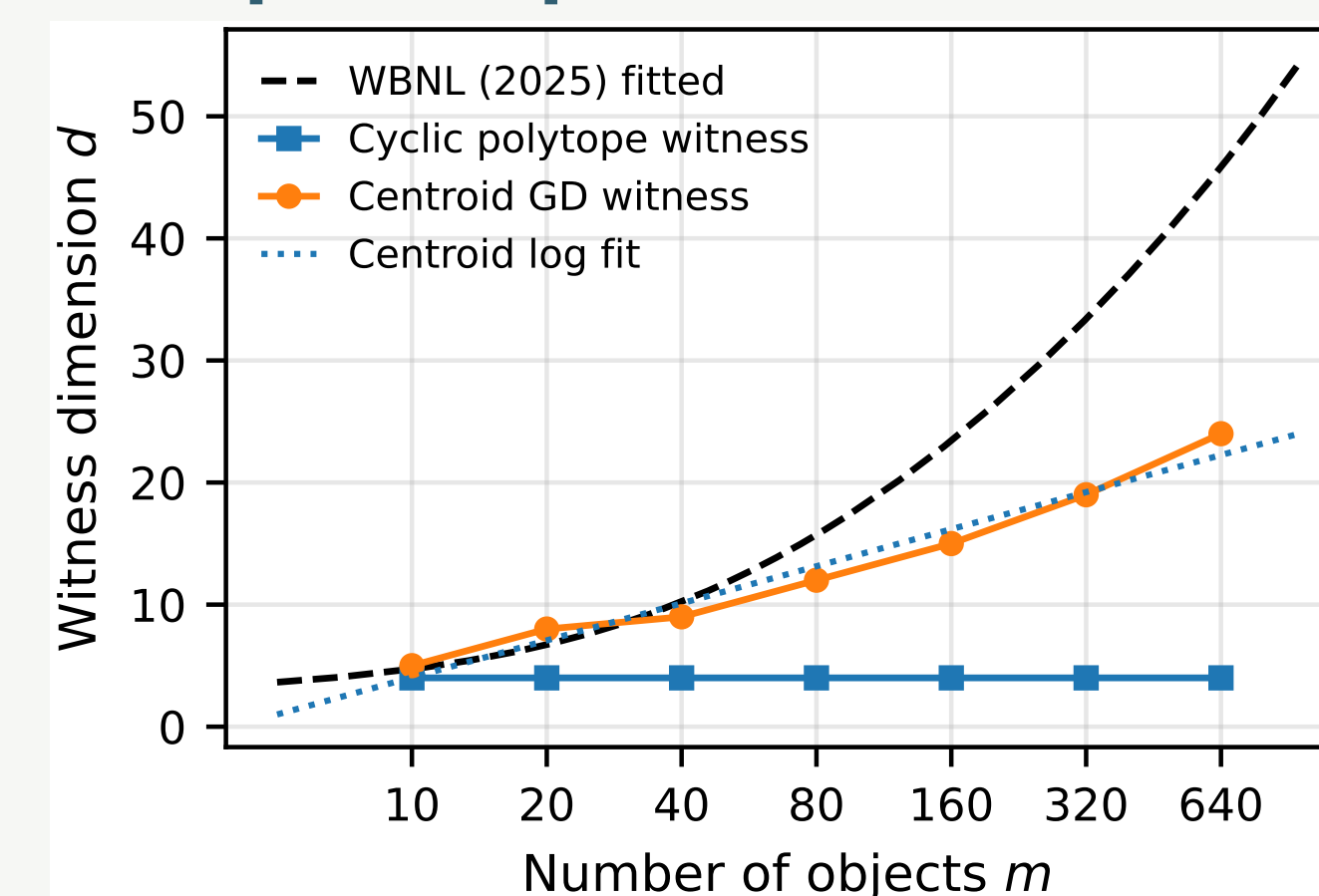
Cosine agrees with inner product after normalization; Euclidean squared-distance comparisons differ by $2\langle u_S, v_i - v_j \rangle$.

Robustness reintroduces m , but the Gaussian witness pays only $\log m$.

3. Empirical

Current “hard” datasets are easy to crack by explicit low-dimensional witnesses.

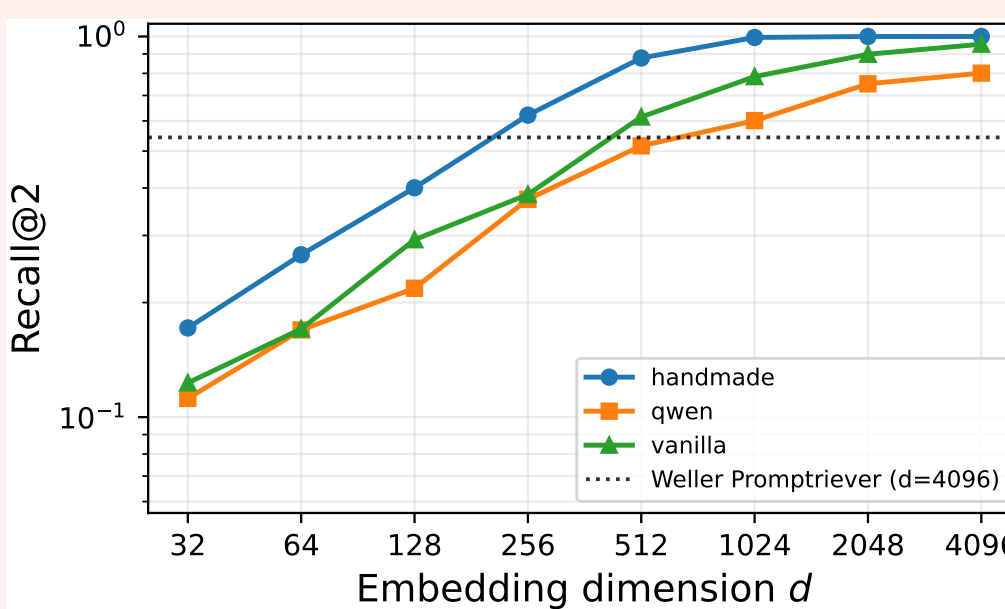
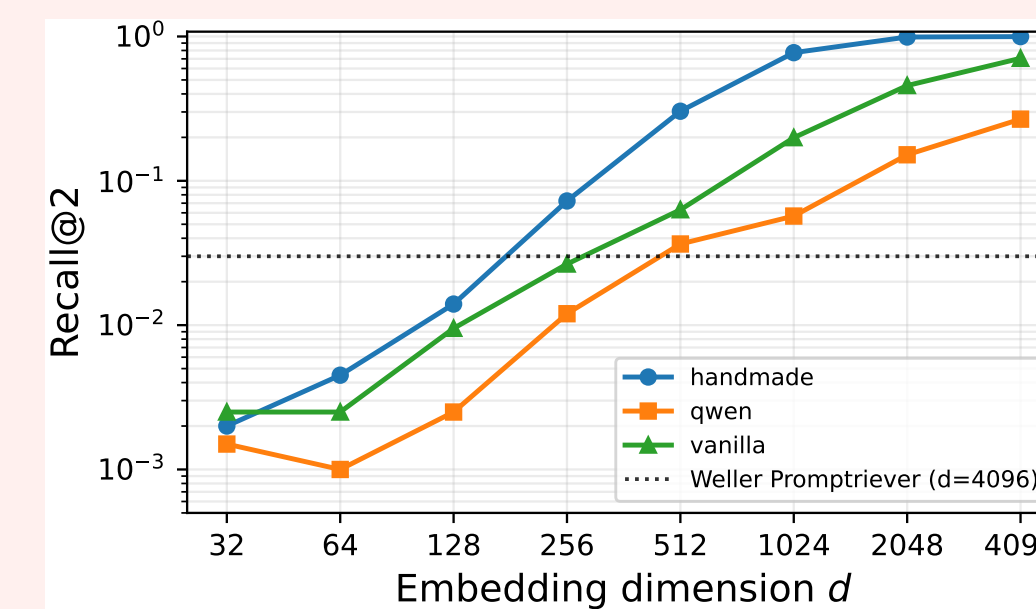
Synthetic top-2: explicit $d = 4$



m	cyclic	MED-C
10	4	5
20	4	8
40	4	9
80	4	12
160	4	15
320	4	19
640	4	24

At $m = 640$, the cyclic construction checks 205120/205120 singleton/pair queries.

LIMIT: random single-vector controls pass



Dataset	baseline	first vanilla	d vanilla	R@2 @4096
LIMIT	0.030	512		0.7060
LIMIT-small	0.543	512		0.9545

Vanilla token sums use no labels and no handcrafted phrases, yet exceed the strongest comparable single-vector Promptriever line. Handmade reaches R@2 0.9980/1.0000 at $d = 4096$.

LIMIT also has an exact $\mathbb{R}^4$ overfit

Dataset	docs	queries	min d	R@2
LIMIT-small	46	1000	4	1
LIMIT	50000	1000	4	1

Takeaway: these failures stress learning, tokenization, conditioning, and optimization, not exact geometric capacity.